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PROFILE

Engineering, product, and program professional with Mercedes-Benz ADAS platform experience and Johns Hopkins Engineering Management training, combining embedded systems integration, technical program execution, AI/product development, data engineering, operations, market research, and commercialization. Built and shipped software, data, AI, and decision-support systems; improved validation coverage to 84%, reduced post-release defects by 26%, raised C++ module test coverage to 95%, supported 75+ internal users, and delivered products reaching 95% adoption. Strong fit across technical product management, AI product, technical program management, systems/integration, data/analytics, product operations, solutions, strategy, and early-career technical roles.

CORE CAPABILITIES

Product / AI / Strategy: discovery, PRDs, roadmaps, prioritization, AI evaluation, human-in-the-loop systems, trust and safety, KPIs, GTM, pricing, market research, commercialization.

Program / Operations: milestones, dependencies, technical risk, release readiness, governance, process mapping, capacity planning, portfolio tracking, executive reporting, cross-functional execution.

Engineering / Data: Embedded C/C++, AUTOSAR, AURIX, HIL/SIL, automotive networks and diagnostics, Python, SQL, REST APIs, ETL/ELT, PostgreSQL, Docker, analytics, automated testing, CI/CD.

EDUCATION

Johns Hopkins University, Whiting School of Engineering, Baltimore, MD

Aug 2025 – Dec 2026

M.S. Engineering Management, GPA: 3.82/4.00

Selected coursework: Deep Learning and Generative AI, Artificial Intelligence, Data Engineering Principles and Practice, AI Project Management, Demand Discovery, Agile Software Development, Accounting and Finance, Innovation and Growth, Leading Change, Professional Presentations

JSS Science and Technology University (JSSSTU), Mysuru, India

Aug 2019 – Jul 2023

B.E. Computer Science and Engineering, GPA: 8.97/10

Selected coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, DBMS, Microprocessors, Software Project Management, Pattern Recognition, Data Mining, Cryptography, Advanced Computer Architecture

PROFESSIONAL EXPERIENCE

Johns Hopkins Technology Ventures, Baltimore, MD

Jun 2026 – Present

Corporate Partnerships Intern

- Map Johns Hopkins research capabilities to industry priorities across emerging life sciences, translating technical research strengths into business-facing insights for corporate partnership and commercialization opportunities.
- Conduct market and technology landscape research to identify partnership themes, commercialization pathways, and areas of strategic fit between university innovation and external organizations.

Mercedes-Benz Research and Development India (MBRDI), Bengaluru, India

Aug 2023 – Feb 2025

ADAS Software Integration / Product Development Engineer - IDC Gen6 Platform Software

- Integrated and validated AUTOSAR-based ADAS software across 5+ vehicle platforms, coordinating architecture, platform, diagnostics, hardware, validation, and vehicle-integration teams around milestones, dependencies, release criteria, and technical risks.
- Configured AUTOSAR Classic BSW, SWCs/CDDs, RTE, communication, diagnostics, and memory stacks; supported AURIX TC39x/TC4D ECU bring-up, NVIDIA Orin/Thor compute integration, Embedded Linux, MCU-MPU IPC/SPI, multicore partitioning, scheduling, watchdogs, and timing validation.
- Performed ECU extract, SIP, DEXT, Next Integration, ECO extract, flashing, readouts, and diagnostic workflows; debugged CAN/CAN-FD, LIN, FlexRay, Ethernet, SOME/IP, DoIP, UDS, DCM, and DEM behavior.
- Executed smoke, regression, HIL, SIL, cluster-HIL, replay, and vehicle validation using CANoe, CANalyzer, CAPL, vTestStudio, DTS Monaco, TRACE32, JTAG, Wireshark, and Linux logs.
- Built readiness dashboards and quality gates across 100+ automated regression tests, improving reported validation coverage to 84% and supporting clearer release go/no-go decisions.
- Standardized defect taxonomy, triage, escalation, acceptance criteria, and runbooks; contributed to a 26% reduction in post-release defects, saved 6+ hours per validation cycle, and improved shipment cadence by 2 days.

Mercedes-Benz Research and Development India (MBRDI), Bengaluru, India

Apr 2023 – Jul 2023

Software Integration Intern

- Built C++ unit-test and code-coverage workflows using GoogleTest, Bazel, gcov, lcov, and genhtml, increasing module coverage to 95% before system and vehicle testing.
- Created mock dependencies, fixed build and compiler-instrumentation issues, wrote targeted tests for uncovered paths, and contributed to CMake-to-Bazel migration and repeatable CI/CD validation workflows.

News Technology Services Pvt Ltd (NewsCorp), Bengaluru, India

Sep 2021 – Nov 2021

Product Development / Backend and Data Engineering Intern

- Built Python/Flask backend services and automated CSV-to-database workflows for 75+ users; added validation, logging, and normalized schemas that reduced structured-data retrieval errors by 30%.
- Automated 5 ETL workflows and improved recurring reporting performance by 26%; supported analytics-ready data delivery with maintainable processing and handoff documentation.

- Conducted stakeholder discovery, translated workflows into requirements and acceptance criteria, prioritized 5+ enhancements, and supported rollout that reached 95% adoption within one month.

Present Technologies (IToWIN), IPN Incubator Center, Portugal

Jan 2026

Product Strategist and Market Research Analyst

- Mapped the U.S. soccer technology ecosystem across 6 competitive segments, evaluated 15+ tools, identified workflow gaps and adoption barriers, and defined 3 core user personas with decision criteria and current tool stacks.
- Synthesized competitive research, stakeholder needs, positioning, GTM themes, pricing hypotheses, and roadmap recommendations into an executive-ready market report for leadership.

FLAGSHIP PORTFOLIO BUILDS

CareerForger - Human-Approved AI Job Application System

- Built a local-first Python/SQLite system that routes JDs across 6 resume families, retrieves evidence, generates bounded AI tailoring tickets, compiles LaTeX resumes, runs deterministic truth/format/keyword QA, packages applications, and supports guarded Manifest V3 autofill with 18 regression tests and no automated submission.

TrustShield AI - Trust and Safety Decision Console

- Built a React/Vite moderation prototype with 8 risk signals, explainable scoring, human-in-the-loop review, policy simulation, network intelligence, analytics, and active-learning concepts; defined precision, false-positive, and reviewer-efficiency targets using synthetic data.

Campus Atlas - AI-Native Higher-Education Operating System

- Built a React/Express prototype and product strategy for a citation-first campus copilot with confidence-gated escalation and admin intelligence; created an offline eval harness that improved scripted answer/escalation/citation/crisis tests from 10/12 to 12/12 passing.

LabOps Intelligence - Engineering Portfolio Planning Simulation

- Built a synthetic decision-support system connecting 24-month demand forecasting, 8 labs, 12 concurrent projects, capacity, budgets, risks, approvals, scenario analysis, and executive triage for program and infrastructure planning.

Cross-Continental Eco-Economic Data Platform

- Built a Python/PostgreSQL/Flask/Docker platform integrating 4 public sources across 5 countries into an analysis-ready monthly dataset with 7 REST endpoints, modular connectors, data-quality controls, cron refresh, and smoke tests.

Lead Routing Copilot - Sales Operations Decision System

- Built an auditable FastAPI intake and routing system where AI proposes intent but versioned rules decide scores, queues, and release; uncertain cases fail closed to human review with rationale, audit trail, threshold simulation, operations console, and 47 automated tests.

SELECTED PRODUCT, STRATEGY, AND ANALYTICS WORK

Bloomberg Children's Center, Johns Hopkins Hospital - Research and Strategy Consulting

- Conducted 10+ stakeholder interviews, mapped intake and registration workflows, benchmarked peer hospitals, defined 5+ operating metrics, and recommended process changes; project reporting showed 15% fewer delays and 22% higher family satisfaction.

KaraOrchee - AI Music Education Product Strategy

- Ran 20+ customer interviews and 34 surveys, compared 11 competitors, validated 85% willingness to pay at \$10–\$20/month, and built pricing, unit-economics, positioning, roadmap, and phased GTM models including a modeled 9:1 LTV:CAC ratio.

NviGo - International Student Services Platform

- Led discovery through 20+ interviews across 5 countries, mapped 8 critical pain points, prioritized the roadmap using severity, fit, and viability, and selected housing as Phase 1 with CAC, retention, and NPS targets.

Stroke Risk Prediction - Machine Learning

- Built and evaluated KNN, Naive Bayes, and ANN models on 5K+ healthcare records using preprocessing, class balancing, scaling, precision/recall analysis, and comparative model evaluation.

LEADERSHIP, TEACHING, AND COMMUNICATION

Teaching Assistant, Johns Hopkins Center for Leadership Education - Oral Presentations; Negotiation and Conflict Resolution, Spring 2026. · **TEDxJHU Speaker** - “Life Is Not a Straight Line,” Nov 2025.

Bureau Club, JSSSTU - Social Media Head - 40% engagement growth. · **Linux Campus Club - Student Coordinator** - technical events and student programs. · **Team Photon - Video Lead** - storyboarding, filming, editing, and event media.

TECHNICAL SKILLS

Programming: Python, C, C++, Embedded C/C++, SQL, CAPL, MATLAB, Java, JavaScript, Solidity

AI / Data: scikit-learn, TensorFlow, Keras, model evaluation, RAG concepts, human-in-the-loop design, REST APIs, Flask, FastAPI, ETL/ELT, PostgreSQL, relational DBMS, data validation, analytics pipelines

Embedded / Automotive: AUTOSAR Classic, SWC, CDD, RTE, BSW, MCAL, RTOS, multicore, AURIX TC39x/TC4D, NVIDIA Orin/Thor, Embedded Linux, CAN/CAN-FD, LIN, FlexRay, Ethernet, SOME/IP, DoIP, UDS, DCM, DEM, IPC, SPI

Testing / Debug: HIL, SIL, vehicle/replay testing, GoogleTest, vTestStudio, CANoe, CANalyzer, CANape, TRACE32, JTAG, Wireshark, DTS Monaco, gcov/lcov, regression testing, defect triage, root-cause analysis

Build / DevOps: Bazel, CMake, Git/GitHub, Jenkins, GitLab, Docker, CI/CD, Coverity, Jira, Confluence

Product / Program: product discovery, PRDs, user stories, roadmap planning, prioritization, KPI/OKR tracking, program planning, milestone/dependency/risk management, release readiness, process mapping, market research, competitive analysis, pricing, GTM, technical documentation

Business Tools: Excel, PowerPoint, Word, Google Workspace

Selected Training: Explore ML (Beginner, Intermediate), Data and Analytics for Business Insights, Speaking Confidently and Effectively, Web Scraping, IoT, Image Processing, SIMTool (IEEE-SJCE)